

COL362/632 Introduction to Database Management Systems

Database Design – Functional Dependencies

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Example

- ▶ Consider the following relation

ActorFilm						
id	firstname	lastname	dob	title	year	language
1	Priyanka	Chopra	1992	Don	2006	Hindi
1	Priyanka	Chopra	1992	Don-II	2011	Hindi
2	Anthony	Hopkins	1937	MI-IV	2011	English
2	Anthony	Hopkins	1937	Valkyrie	2008	English
3	Bill	Nighty	1949	Valkyrie	2008	English

- ▶ Problems with this design

1. Redundancy
2. Update Anomalies

- ▶ `update ActorFilm set firstname = 'Priyankaa' where id = 1;`
- ▶ `insert into ActorFilm values (null, null, null, null, Avatar, 2009, English);`
- ▶ `delete from ActorFilm where title = 'Valkyrie';`
- ▶ `delete from ActorFilm where firstname = 'Anthony' and lastname = 'Hopkins';`

Normalization

The process of systematically eliminating redundancy and anomalies.

Functional Dependencies

- ▶ A **functional dependency** is another kind of constraint
- ▶ If two tuples in a relation agree on the values of one set of attributes, then they must also agree on the values of another set of attributes

Given two attribute sets A and B with $A, B \subseteq [R]$

$A \rightarrow B$ is called functional dependency (FD) if a function exists $f(a) := b \quad \forall a \in \pi_A(R), b \in \pi_B(R)$ for all possible instances of R .

- ▶ In other words, values of A determine the values of B

Functional Dependencies

Example

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Functional Dependencies (FDs)

- $\{ \text{id, firstname, lastname} \} \rightarrow \{ \text{dob} \}$
- $\{ \text{title} \} \rightarrow \{ \text{language} \}$
- $\{ \text{firstname, lastname} \} \rightarrow \{ \text{title, year} \}$

Incorrect FD!

Functional Dependencies

- ▶ Important to differentiate between instance and schema
- ▶ FD defines a constraint for the possible instances of R
 - Instances that satisfy FD are called **legal instances**
- ▶ FD do not define dependencies for concrete instance
 - Single counter example is enough!

Functional Dependencies

Example

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Functional Dependencies

- $\{ \text{id} \} \rightarrow \{ \text{firstname}, \text{lastname}, \text{dob} \}$
- $\{ \text{firstname} \} \rightarrow \{ \text{lastname} \}$ **not an FD**
- $\{ \text{dob} \} \rightarrow \{ \text{id}, \text{firstname}, \text{lastname} \}$ **not an FD**

Functional Dependencies

- ▶ Not just a theoretical DB concept!
- ▶ Play a crucial role in applications beyond DB design
 - **Feature engineering and selection in ML** (e.g., dimensionality reduction, feature generation)
 - **Data cleaning and anomaly detection** (e.g., missing value imputation, duplicate detection, data errors)
 - **Causal discovery and data understanding in ML**
 - **Query Optimization in Data pipelines**
 - **Data integration and schema matching**
 - **Balancing data privacy and data utility**

Super Keys

- ▶ A **superkey** is a set of one or more attributes (columns) in a relation (table) such that the values of these attributes are sufficient to uniquely identify a tuple (row) in the relation.
- ▶ Any subset $S \subseteq [R]$ for which $S \rightarrow [R]$ holds is called a **super key** of R
 - $[R] \rightarrow [R]$
 - $\exists a \in S : S \setminus \{a\} \rightarrow [R]$
- ▶ Super key functionally determines all other attributes
 - Not necessarily minimal!
- ▶ **Example**
 - ActorFilm(id, firstname, lastname, dob, title, year, language)
 - $S = \{ \text{id, firstname, lastname, dob, title} \}$

Candidate Key

- ▶ A **candidate key** is a **minimal superkey**
- ▶ For a superkey $S \subseteq [R]$ for which
 - $\nexists a \in S : S \setminus \{a\} \rightarrow [R]$ or
 - $\forall a \in S : S \setminus \{a\} \nrightarrow [R]$
 - is called the **candidate key** of R
- ▶ **Example**
 - ActorFilm(id, firstname, lastname, dob, title, year, language)
 - $S = \{ \text{firstname, lastname, title, year} \}$
- ▶ A relation can have **multiple candidate keys**

Primary Attribute & Primary Key

- ▶ Let C be the set of all candidate keys of $[R]$, then
- ▶ all attributes $\bigcup_i c_i$ are called **primary attributes** of $[R]$

Primary Key

- ▶ Any candidate key used to identify a tuple!
 - Cannot have null
 - Not change over time
 - Short as possible

Inferring FDs

- ▶ Given a set of FDs, which other FDs follow from it?
- ▶ Example
 - Given: $R(A, B, C)$ satisfies $A \rightarrow B$ and $B \rightarrow C$
 - Inferred: $A \rightarrow C$ through **transitivity of FDs**.
 - Given: $\{\text{aadhar}\} \rightarrow \{\text{pan}\}, \{\text{pan}\} \rightarrow \{\text{dob}\}$
 - Then, $\{\text{aadhar}\} \rightarrow \{\text{dob}\}$

Inferring FDs

- ▶ Given: $A_1A_2 \dots A_n \rightarrow B_1B_2 \dots B_m$
- ▶ Inferred $A_1A_2 \dots A_n \rightarrow B_i$ for $i = 1, 2, \dots, m$ through **splitting rule**

- ▶ Given $A_1A_2 \dots A_n \rightarrow B_i$ for $i = 1, 2, \dots, m$
- ▶ Inferred: $A_1A_2 \dots A_n \rightarrow B_1B_2 \dots B_m$ through **combining rule**

Equivalent FDs

- ▶ Two FDs S and T are equivalent if
 - set of relation instances satisfying S is exactly the same as the set of relation instances satisfying T
- ▶ If FD S follows from FD T , if every instance that satisfies T also satisfies S
- ▶ $S = T$ iff $S \implies T$ and $T \implies S$

Trivial and Non-trivial FDs

- ▶ Consider $R(A, B, C)$
- ▶ Given FDs
 - $AB \rightarrow A$ trivial
 - $AB \rightarrow AC$ non-trivial
 - $AB \rightarrow C$ completely non-trivial
- ▶ **Trivial FD** $X \rightarrow Y$ where $Y \subseteq X$
- ▶ **Non-trivial FD** $X \rightarrow Y$ where $Y \not\subseteq X$
- ▶ **Completely non-trivial FD** $X \rightarrow Y$ where $Y \cap X = \emptyset$

Armstrong's Axioms

- | | | | | |
|----|-------------------------------------|------------|------------------------------------|----------------------------------|
| 1. | $B \subseteq A$ | $\implies$ | $A \rightarrow B$ | Reflexivity |
| 2. | $A \rightarrow B$ | $\implies$ | $AC \rightarrow BC$ | Augmentation |
| 3. | $A \rightarrow B, B \rightarrow C$ | $\implies$ | $A \rightarrow C$ | Transitivity |
| 4. | $A \rightarrow BC$ | $\implies$ | $A \rightarrow B, A \rightarrow C$ | Decomposition (splitting) |
| 5. | $A \rightarrow B, A \rightarrow C$ | $\implies$ | $A \rightarrow BC$ | Union (combining) |
| 6. | $A \rightarrow B, DB \rightarrow C$ | $\implies$ | $DA \rightarrow C$ | Pseudo-transitivity |

Closure of FDs

- ▶ Given: the set S of FDs
- ▶ Output: S^+ , the **closure** of S containing all FDs that can be derived from S
- ▶ Example
 - Given $S = \{ AB \rightarrow C, C \rightarrow ED \}$
 - Then $S^+ = \{ AB \rightarrow ED, AB \rightarrow E, AB \rightarrow D, ABC \rightarrow ED, C \rightarrow E, C \rightarrow D, \dots \}$
 - HW: Find all FDs

Closure of Attributes

- ▶ Let S be a set of FDs
 - Is a new FD F derivable from S ?
- ▶ Example
 - Given
$$S = \{ AB \rightarrow C, BC \rightarrow AD, D \rightarrow E, CF \rightarrow B \}$$
 - Does the FD $AB \rightarrow D$ hold?
 - Compute $\{A, B\}^+$

FD	$\{A, B\}^+$
$AB \rightarrow C$	$\{A, B, C\}$
$BC \rightarrow AD$	$\{A, B, C, D\}$
$D \rightarrow E$	$\{A, B, C, D, E\}$

- $AB \rightarrow D$ holds!

Attribute Closure Method

Given

- ▶ set S of FDs
- ▶ candidate FD $F : A \rightarrow B$

Step 1: Compute A^+

$A^+ = A$ initial set of attributes

```
do {  
  for each FD  $X \rightarrow Y \in S$  {  
    if  $X \subseteq A^+$  then  $A^+ \leftarrow A^+ \cup Y$   
  }  
  if  $A^+$  unchanged, quit  
}
```

Step 2: Check if $B \subseteq A^+$